

House Price Prediction: Summary

Introduction

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property features such as area, number of bedrooms, bathrooms, stories, parking availability, furnishing status, and other amenities. The project involved data preprocessing, exploratory data analysis, visualization, model building, performance evaluation, and interpretation of the results.

Data Preparation

The Housing dataset was loaded using the Pandas library. The dataset contained no missing values, which eliminated the need for missing value treatment. Duplicate records were checked and removed where necessary. Since several columns contained categorical values such as "yes/no" and furnishing status, they were converted into numerical values using one-hot encoding so that they could be used by machine learning algorithms.

Model Development

The dataset was divided into training and testing sets using an 80:20 ratio. Two regression models were developed and evaluated: Linear Regression and Random Forest Regressor. Their performances were compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 Score. The evaluation results showed that the Linear Regression model performed better than the Random Forest model on this dataset. The Linear Regression model achieved an R^2 Score of approximately **0.653**, indicating that it explained about **65% of the variation in house prices**.

Data Visualization and Insights

Several visualizations were created to better understand the data. The histogram showed that house prices are positively skewed, with most houses falling within the lower to middle price range and only a few properties having very high prices. The correlation heatmap revealed that **area**, **bathrooms**, **air conditioning**, **stories**, **parking**, and **bedrooms** have the strongest positive relationship with house price. The feature importance analysis further confirmed that **area** is the most influential feature affecting house prices, followed by bathrooms, air conditioning, parking availability, stories, and bedrooms.

Conclusion

The project demonstrates that machine learning techniques can effectively predict house prices using property characteristics. Among the models tested, Linear Regression provided the best overall performance for this dataset. The findings suggest that house area and essential amenities play a significant role in determining property value. Real estate companies can use these insights to improve pricing strategies, identify valuable property features, and make more informed decisions while buying, selling, or valuing residential properties.